

Supplementary information

Artificial Neural Networks for Predicting Optical Conversion Efficiency in Luminescent Solar Concentrators

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SI.1. Optical performance quantification

SI.1.1 Optical conversion efficiency (η_{opt})

In a luminescent solar concentrator, the optical conversion efficiency (η_{opt}) is given by the ratio between the output optical power (P_{out}) and the input optical power (P_{in}). Regarding the experimental assessment of η_{opt} , we note that different expressions are presented in the literature, as outlined below.

The most often used quantification of η_{opt} is through equations S1 and S2, namely [1,2]:

$$\eta_{opt} = \frac{P_{out}}{P_{in}} = \frac{I_{SC}^L V_0^L A_e \eta_{solar}}{I_{SC} V_0 A_s \eta_{PV}} \quad (S1)$$

where I_{SC}^L and V_0^L represent the short-circuit current and the open circuit voltage when the PV device is coupled to the LSC (I_{SC} and V_0 are the corresponding values of the PV device exposed directly to the solar radiation), η_{solar} is the efficiency of the PV device relatively to the total solar spectrum and η_{PV} is the efficiency of the PV device at the LSC emission wavelengths.

An alternative definition is also often reported given by [3]:

$$\eta_{opt} = \frac{I_{SC}^L A_e}{I_{SC} A_s} \quad (S2)$$

Another experimental approach used to quantify the optical conversion efficiency, considers that η_{opt} can be described by weighting all the losses in the LSC, given by the product of several terms [4]:

$$\eta_{opt} = (1 - R) \eta_{abs} \eta_{SA} \eta_{yield} \eta_{Stokes} \eta_{trap} \eta_{mat} \quad (S3)$$

in which:

- $R = (1 - n_i)^2 / (1 + n_i)^2$ is the Fresnel reflection coefficient for perpendicular incidence, in

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which n_i represents the refractive index of the emitting medium at the incident wavelength (λ_i).

- $\eta_{abs}=1-10^{-A}$ is the ratio of photons absorbed by the emitting layer to the number of photons falling on it, with A representing the absorbance value at λ_i .
- η_{SA} is the self-absorption efficiency, arising from self-absorption of the emitting centres, as detailed elsewhere [5].
- η_{yield} is the emission quantum yield of the optically active centre at λ_i .
- $\eta_{Stokes}=\lambda_i/\lambda_p$, is the Stokes efficiency calculated by the energetic ratio between the average energy of the emitted photons and the incident energy.
- η_{tr} takes into account the transport losses due to matrix absorption and scattering, frequently it is considered that $\eta_{tr}=1$, as the transport and scattering losses are neglected.
- $\eta_t=(1-1/n_p^2)^{1/2}$ is the trapping efficiency, where n_p is the refractive index of the emitting medium at λ_p , and is defined as the fraction of photons confined within the substrate, i.e., the fraction of emitted photons escaped from the edge versus the total emitted photons. This term accounts for the emission losses at the surface through a so-called escape cone with an aperture angle defined by $\theta_c=2\times\sin^{-1}(1/n_p)$.

SI.1.2 Absolute emission quantum yield (η_{yield})

The emission quantum yield (ϕ) is given by equation S4 [6]:

$$\eta_{yield} = \frac{N_{Emi}}{N_{Abs}} = \frac{\int_{\lambda_1}^{\lambda_2} \frac{\lambda}{hc} [I_{em}^{sample}(\lambda) - I_{em}^{reference}(\lambda)] d\lambda}{\int_{\lambda_3}^{\lambda_4} \frac{\lambda}{hc} [I_{ex}^{sample}(\lambda) - I_{ex}^{reference}(\lambda)] d\lambda} \quad (S4)$$

where N_{Abs} and N_{Em} are the numbers of photons absorbed and emitted by a sample, respectively, h is Planck's constant, c is the speed of light, I_{em}^{sample} and $I_{em}^{reference}$ are the emission intensities measured with and without a sample respectively, in the emission spectra wavelength interval $[\lambda_1, \lambda_2]$ and I_{ex}^{sample} and $I_{ex}^{reference}$ are the integrated

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intensities of the excitation radiation measured with and without a sample respectively, in the excitation wavelength interval $[\lambda_3, \lambda_4]$. In order to validate the calculation methodology, the emission quantum yield of a 1 mm thick pellet of the white standard sodium salicylate (Merck P.A.) and of the fluorescent plastic standards provided by the manufacturer with known emission quantum yield values was also measured under the same experimental conditions. The reproducibility of the measurement was tested by performing three measurements for each sample revealing a standard deviation below 0.5 %. The measurement accuracy is within 10% according to the manufacturer.

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SI.2.-Dataset

Table S1. The post processing dataset used in this work, obtained in May 2023 from the public repository [7].

Entry #	Materials				General Optical Features							Target Optical Feature
	Optical center type	Host		Processing	Absorption			Emission			Quantum Yield	
		Type	Chemical		A_p (nm)	A_{min} (nm)	A_{max} (nm)	E_p (nm)	E_{min} (nm)	E_{max} (nm)	η_{yield} (%)	η_{opt} (%)
1	dye	hybrid	GLYMO/TMSO/ASB	film	578	420	600	613	550	750	98	18.80
2	QD	polymer	PMMA	bulk	730	600	900	910	800	1000		2.50
3	QD	polymer	PMMA	bulk	730	600	900	910	800	1000		2.50
4	QD	polymer	PMMA	fiber	730	600	900	910	800	1000		4.00
5	QD	polymer	PMMA	fiber	730	600	900	910	800	1000		7.00
6	QD	solvent	toluene	solution	800	400	800	900	700	1000	30	1.40
7	QD	solvent	toluene	solution	600	400	600	630	600	700	50	0.50
8	QD	solvent	toluene	solution	376	300	650	623	600	700	50	0.30
9	dye	solvent	chlorobenzene	solution	413	400	600	657	600	800	67	3.40
10	dye	solvent	PC/EG	solution	550	450	600	629	550	750	95	2.60
11	dye	solvent	PC/EG	solution	466	300	750	685	700	900	11	0.60
12	dye	solvent	PC/EG	solution	557	300	750	802	700	900	9	0.50
13	polymer	solvent	chlorobenzene	solution	460	350	550	592	550	800	45	1.00
14	dye	polymer	PCMA	film	403	300	450	471	400	600	40.8	7.70
15	dye	hybrid	d-U(600)	bulk	370	300	450	650	450	750	67	5.50
16	NP	polymer	PMMA	bulk	470	400	550	597	500	725		4.09
17	dye	polymer	PMMA	bulk	340	300	400	480	400	600	78	0.40
18	NP	polymer	PMMA/PLMA	bulk	375	300	450	750	400	900	45	4.25
19	Lndye	polymer	PMMA	fiber	540	350	550	630	500	700		0.29
20	NP	polymer	PMMA	bulk	550	400	750	800	600	1000	80	6.80
21	QD	polymer	PLMA	bulk	650	500	770	718	520	850	63	1.75

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22	QD	solvent	OLA/DDT	solution	650	500	770	718	520	850	63	3.67
23	CD	polymer	PMMA	bulk	350	300	400	435	380	575		12.23
24	CD	hybrid	liquid crystal	bulk	350	280	400	450	390	600		4.52
25	CD	hybrid	KH-792	film	350	300	425	510	400	650	45	12.00
26	CD	polymer	PVP	film	325	300	400	430	380	600		5.02
27	CD	polymer	PVP	bulk	340	300	600	540	450	700	40	0.92
28	CDQD	polymer	PVP; PLMA-co-EGDA	bulk	400	300	500	500	400	550	70	3.05
29	NP	polymer	PVP	film	400	300	500	600	500	700	25	1.85
30	NP	polymer	polystyrene	film	325	300	520	518	475	525	58	2.40
31	CD	polymer	PMMA	film	380	280	700	420	400	650		4.75
32	CD	polymer	PLMA	bulk	440	300	500	540	450	700	25	1.20
33	QD	polymer	PLMA	bulk	500	400	600	830	600	1000	50	2.85
34	QD	polymer	acrylate nanocomposite	film	450	350	750	862	800	1100	91	8.10
35	QD	polymer	PMMA	film	350	400	800	740	600	850		6.97
36	QD	polymer	PVP	bulk	500	350	650	805	600	1000	78	6.40
37	QD	polymer	PLMA	bulk	640	400	1000	960	700	1200	40	3.27
38	QD	polymer	PMMA	bulk	415	310	620	918	620	1240	60.3	3.94
39	Ln	Inorganic	CaI2/NaCl	film	650	400	900	1140	1000	1200	100	14.90
40	dye	polymer	PDMS	bulk	478	250	600	601	550	800	61.1	22.00
41	dye	polymer	PDMS	bulk	513	300	650	642	650	800	24.8	3.30
42	dye	polymer	PDMS	bulk	449	250	550	571	500	800	80	27.80
43	dye	polymer	PDMS	bulk	473	250	600	607	500	800	44.3	24.70
44	QD	polymer	OSTE	bulk	450	350	1000	825	650	1000	55	7.90
45	dye	polymer	PMMA	bulk	491	300	500	581	550	650	95	23.70
46	Ln	polymer	Parylene	film	347	250	400	613	570	710		2.47
47	Ln	polymer	photopolymerizable composition "Prism" (PPC)	film	590	280	600	630	550	700	65	11.30
48	Ln	polymer	PMMA	film	380	250	400	612	570	710	30.5	0.34
49	Ln	hybrid	t-U(5000)	film	370	290	380	611	570	710	27	3.20
50	Ln	polymer	PVA	film	290	280	400	546	380	650	40	8.80
51	Ln	hybrid	BS	film	350	240	420	612	570	710	34	4.30

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52	Ln	hybrid	4,40-diureido-2,20-bipyridine bridged organosilane	film	325	240	400	612	570	710	8	1.20
53	Ln	hybrid	4,40-diureido-2,20-bipyridine bridged organosilane	film	325	240	400	544	450	650	12	1.70
54	Ln	hybrid	t-U(5000)	film	370	240	400	612	570	710	63	9.00
55	Ln	hybrid	d-U(600)	film	370	300	380	615	570	710	61	1.20
56	Ln	hybrid	tetracarboxamide-BS	film	380	240	380	612	570	710	23	0.43
57	Ln	hybrid	d-U(600)	film	360	240	380	612	570	710	30	0.01
58	Ln	hybrid	t-U(5000)	fiber	360	240	420	615	570	710	85	2.30
59	dye	hybrid	d-U(600)	film	520	300	540	590	550	700	78	0.08
60	dye	hybrid	d-U(600)	fiber	520	300	540	590	550	700	78	1.60
61	dye	hybrid	t-U(5000)	fiber	520	300	540	590	550	700	93	8.00
62	dye	hybrid	t-U(5000)	film	770	300	800	775	700	950	16	1.50
63	dye	hybrid	d-U(600)	bulk	665	250	700	670	600	750	12	3.70
64	Ln	hybrid	t-U(5000)	fiber	370	300	450	615	570	710	89	0.70
65	dye	hybrid	t-U(5000)	fiber	560	300	600	580	550	700	95	2.10
66	dye	hybrid	t-U(5000)	fiber	780	300	750	730	650	850	21	0.50
67	QD	polymer	PMMA	bulk	473	350	550	640	600	700	45	1.00
68	QD	polymer	PMMA	bulk	350	300	450	550	450	750	56	26.50
69	QD	polymer	PMMA	bulk	470	400	600	550	450	700	20	2.01
70	CD	hybrid	silicone dendrimers	film	450	310	510	525	500	700	11	7.58
71	dye	polymer	acrylamide	film	545	400	700	680	550	800	50	12.50
72	dye	solvent	water	solution	498	300	700	580	520	700	30	6.88
73	dye	solvent	water	solution	569	350	600	595	550	750	61	2.58
74	dye	solvent	water	solution	488	300	500	510	450	600	51	3.30
75	CD	polymer	PNIPAm	bulk	460	350	500	510	350	500	54	2.70
76	CD	polymer	PVP	film	420	300	550	515	400	700	40	1.60
77	CD	polymer	PVA; PVP; PVP+PEI	film	557	300	600	612	400	700	70	2.30
78	CD	polymer	PVP	film	420	300	550	500	440	600	67	2.20
79	CD	solvent	water	solution	491	250	520	520	500	700	82	5.43
80	CD	hybrid	d-U(600)	film	510	250	520	535	400	650	78	0.06

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81	CD	hybrid	d-U(600)	fiber	510	250	520	535	400	650	78	1.70
82	dye	polymer	PMMA	bulk	575	400	610	620	570	720	95	19.00
83	CD	polymer	PVP	film	405	300	500	520	400	700	70	3.20
84	CD	polymer	PVP	film	405	300	500	520	400	700	65	2.90
85	QDdye	polymer	PLMA-co-EGDA	bulk	500	400	700	600	450	750	32.7	0.12
86	CD	polymer	PVP	film	347	280	500	540	450	700	61	4.56
87	Ln	polymer	thiol-ene	bulk	405	250	500	520	450	700	81	3.40
88	CD	polymer	HEMA/EGDMA	film	380	300	450	450	400	700	11.54	1.36
89	QD	polymer	EVA	film	320	300	550	670	600	700	32.97	2.55
90	CDQD	polymer	HEMA/EGDMA/EVA	film	320	300	550	515	400	700	23	1.89
91	CDQD	polymer	HEMA/EGDMA/EVA	film	320	300	550	500	400	700	26	3.76
92	CD	hybrid	silicone resin	film	400	300	500	517	450	700	33	4.50
93	CD	hybrid	silicone resin	bulk	400	300	500	517	450	700	41	5.89
94	CD	hybrid	silicone resin	bulk	400	300	500	517	450	700	41	3.13
95	CD	polymer	PVP	film	555	300	650	650	575	800	17.6	3.00
96	CD	polymer	PVP	film	450	300	660	600	500	800	60	4.30
97	CD	solvent	ethanol	solution	340	300	450	520	400	700		1.23
98	CD	polymer	PVP	film	340	300	450	520	400	700		0.90
99	CD	polymer	PVP	film	380	300	500	570	450	700	41.52	3.51
100	CD	polymer	PVP	film	355	300	500	520	450	650	15.01	2.76
101	CD	polymer	PVP	film	410	300	600	608	550	750	7.6	2.77
102	CD	polymer	PVP	film	400	300	600	600	450	750	22	4.03
103	QD	polymer	thiol-ene	bulk	600	400	650	625	550	700	30	2.70
104	polymer	polymer	PMMA	film	500	392	250	491	650	400	47	5.71
105	polymer	polymer	PMMA	film	500	392	250	473	650	450	57	9.11
106	polymer	polymer	PMMA	film	500	392	250	454	650	400	68	12.08
107	polymer	polymer	PMMA	film	500	392	250	491	650	400	47	5.71
108	polymer	polymer	PMMA	film	500	392	250	473	650	450	57	9.11
109	polymer	polymer	PMMA	film	500	392	250	454	650	400	68	12.08
110	NP	polymer	PLA	bulk	600	511	450	527	580	490	65	0.15
111	dye	polymer	PLA	bulk	600	554	450	612	700	575	70	0.16

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112	QD	solvent	n-hexane	solution	510	450	300	517	550	475	89	2.32
113	dye	solvent	water	solution	660	480	680	665	600	800	31	2.65
114	QD	polymer	polyester	film	400	300	500	630	550	700	67	2.50
115	QD	polymer	PLMA-co-EGDA	film	480	380	550	650	600	700	4.7	4.90
116	CD	polymer	PMMA	film	490	300	600	595	550	700	50	4.81
117	QD	glass	glass	bulk	500	350	525	521	500	550	73	6.17
118	QD	polymer	PMMA	bulk	343	300	750	600	400	850	39.2	0.70
119	QD	polymer	PDMS	bulk	350	300	700	400	380	450	0.7	6.70
120	QD	polymer	PDMS	bulk	400	300	700	600	400	700	4	8.43
121	QD	polymer	PMMA	film	400	300	528	510	485	530		12.26
122	QD	polymer	PMMA	film	314	300	350	578	350	760		1.15
123	QD	solvent	toluene	solution	460	350	500	590	420	700		11.00
124	QD	solvent	anhydrous dimethyl sulfoxide and N-methyl-2-pyrrolidione	film	500	350	780	780	680	830		29.00
125	QD	polymer	polyester PU acrylate oligomer	film	500	400	750	700	625	775	99.4	3.10
126	QD	polymer	thiol-ene	bulk	450	400	650	523	500	575	78	2.11
127	QD	polymer	PMMA	film	374	350	800	710	600	800	56	2.00
128	QD	polymer	PMMA	film	523	300	525	531	500	570	60	5.40
129	QD	polymer	PLMA-co-EGDA	bulk	420	300	650	570	475	700	60	2.00
130	QD	polymer	PMMA	film	450	375	525	520	450	575	70	0.87
131	dye	polymer	starch, sucrose and trehalose	film	580	250	620	620	550	800	75	3.20
132	QD	polymer	poly(cyclohexylethylene)	film	450	350	600	850	600	1000	73	2.50
133	QD	polymer	acrylate-based monomers	film	550	300	750	800	650	1000	70	4.36
134	QD	polymer	PDMS	film	450	350	650	630	550	700	79.5	7.60
135	QD	polymer	PVP	film	400	300	600	590	450	700	34.1	1.01
136	QD	polymer	PMMA	bulk	400	300	600	700	500	900	70	3.80
137	QD	polymer	PVP	film	450	350	600	628	550	700	70	3.00
138	QD	polymer	PLMA-co-EGDA	bulk	500	350	650	625	550	675	40	5.30
139	QDdye	polymer	PLMA-co-EGDA	bulk	590	300	650	600	450	700	32.7	1.00
140	QD	polymer	PLMA-co-EGDA	bulk	700	500	900	900	700	1100	70	1.20
141	QD	polymer	PLMA	bulk	500	400	700	750	600	900		5.70

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142	QD	polymer	PLMA	bulk	450	300	530	620	550	700	54.3	1.79
143	QD	polymer	3-mercaptopropionic acid	film	550	350	700	700	400	1000	45	1.68
144	QD	polymer	PMMA	bulk	470	400	650	560	475	700	20	2.01
145	QD	polymer	PVP	film	450	300	800	630	580	700	50	1.93
146	QDdye	polymer	PVP	film	500	350	600	560	450	700	86	16.32
147	QD	polymer	poly(vinylidene fluoride)	film	350	300	600	580	500	650		0.81
148	QD	polymer	PVP	film	400	300	550	550	450	700	91.1	6.40
149	QD	solvent	KH-792	solution	362	300	450	442	400	650	62	4.40
150	dye	polymer	EHM-co-FMA and BMIH	film	570	300	650	620	550	750	50	4.90
151	dye	polymer	GTPn-based	film	500	400	600	630	500	750	6.6	7.70
152	dye	polymer	PMMA	film	450	300	550	568	450	700	27.2	8.00
153	dye	polymer	PMMA	bulk	420	300	500	550	400	800	35	2.33
154	dye	polymer	PBzMA	film	400	300	450	490	400	700	75	2.32
155	dye	polymer	PDMS	bulk	520	450	550	538	500	700	93	10.18
156	dye	polymer	PDMS	bulk	520	450	550	538	500	700	93	10.12
157	QD	polymer	PMMA	bulk	516	300	550	528	470	750	100	82.20
158	QD	polymer	PMMA	bulk	487	300	600	599	550	650	100	91.80
159	QD	polymer	PMMA	bulk	437	300	450	590	525	700	100	91.80
160	dye	polymer	PMMA	fiber	575	300	610	620	570	720	100	21.00
161	dye				575	300	610	620	570	720	3	2.00
162	dye	polymer		bulk	575	300	610	620	570	720	96	37.73
163	dye	polymer	styrene acrylonitrile resin	bulk	576	300	610	647	575	800	96	8.20
164	dye	polymer	PMMA	bulk	576	300	610	648	575	800	96	8.90
165	dye	polymer	polycarbonate	bulk	576	300	610	651	575	800	96	8.30
166	dye	polymer	polycarbonate	bulk	576	300	610	651	575	800	96	8.00
167	dye	polymer	polycarbonate	bulk	576	300	610	650	575	800	96	8.90
168	dye	polymer	PMMA	bulk	370	300	500	509	480	750	68	5.20
169	dye	polymer	PMMA	bulk	370	300	500	509	480	750	68	3.70
170	QD	polymer	PLMA-co-EGDA	bulk	400	300	550	513	475	550	65	1.22
171	dye	polymer	PMMA	bulk	400	300	550	600	400	800		6.40
172	dye	polymer	PDMS	bulk	461	300	550	590	500	750	100	4.20

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173	CD	polymer	PDMS	film	325	200	700	555	500	700	32	4.26
174	QDNP	polymer	thiol-ene	film	440	410	523	525	440	710	61	2.90
175	Ln	polymer	Disperbyk 2015	film	330	300	400	615	570	640	20	4.30
176	dye	polymer	PVP	film	635	400	700	909	800	1000	12.93	1.21
177	QD	polymer	PMMA	film	390	350	463	478	425	550		2.52
178	QD	polymer	PMMA	film	425	350	520	525	475	550		4.12
179	CD	polymer	PVA	film	366	300	450	450	390	600	57.77	1.98
180	QD	solvent	toluene	film	350	300	450	511	420	775	65.4	3.26
181	QD	solvent	toluene	film	350	300	620	630	450	775	94.6	6.53
182	QD	solvent	toluene	film	350	300	620	570	420	775	80	9.94
183	dye	polymer	polymer	bulk	530	300	600	560	500	700	95	19.00
184	Ln	glass	glass	bulk	583	500	900	1060	1000	1200	62	10.00
185	dye	polymer	PMMA	film	530	300	600	560	500	700		6.30
186	Ln	glass	glass	bulk	583	500	900	1060	1000	1200	68	5.90
187	dye	polymer	PMMA	film	523	400	550	577	520	650	99	20.50
188	dye	polymer	PMMA	film	560	400	700	651	575	850	31	1.40
189	dye	polymer	PMMA	film	542	400	650	633	575	850	38	3.10
190	dye	polymer	PMMA	film	523	400	700	618	575	850	68	4.40
191	dye	polymer	PMMA	film	447	400	525	550	525	700	66	1.80
192	CD	polymer	PVP	film	477	350	600	575	525	700	80	5.17
193	CD	polymer	PLMA-co-EGDA	bulk	300	300	520	460	380	630		6.07

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Table S2. Hold-up test set applied in the ANN model to predict η_{opt} .

Entry #	Materials				General Optical Features								Ref.
	Optical center type	Host		Processing	Absorption			Emission			Quantum Yield	Target Optical Feature	
		type	Chemical		A_p (nm)	A_{min} (nm)	A_{max} (nm)	E_p (nm)	E_{min} (nm)	E_{max} (nm)	η_{yield} (%)	η_{opt} (%)	
1	NPdye	polymer	PLA	bulk	600	530	450	550	700	500	68.0	0.22	[8]
2	dye	polymer	PMMA	bulk	340	300	400	540	400	700	14.0	0.25	[9]
3	Ln	polymer	PMMA	film	360	250	380	545	450	700	1.6	0.27	[10]
5	polymer	solvent	chlorobenzene	solution	467	300	550	593	550	700	48.0	0.90	[3]
6	QD	polymer	polystyrene	film	337	300	500	520	475	575	79.0	1.10	[12]
7	dye	polymer	PMMA	bulk	461	350	550	552	450	700	71.0	1.30	[13]
8	CDQD	polymer	PVP/PLMA	bulk	450	300	550	620	450	700	45.0	1.40	[14]
9	QD	polymer	PDMS	bulk	500	350	750	627	550	700	35.9	1.45	[15]
10	dye	polymer	PMMA	film	436	300	600	562	450	750	42.0	1.60	[16]
11	QD	polymer	thiol monomer, allyl monomer, and photoinitiator	bulk	396	300	400	600	400	700	40.7	2.23	[17]
12	QD	polymer	polystyrene	film	466	300	559	541	500	700	90.0	2.50	[18]
13	CDQD	polymer	HEMA/EGDMA/EVA	film	320	300	550	670	400	700	22.0	2.54	[19]
14	CD	polymer	PVP	film	404	300	550	594	500	750	86.4	2.60	[20]
15	CD	polymer	PDMS	film	350	200	800	585	500	700	21.0	2.83	[21]
16	QD	polymer	Polystyrene	film	480	300	500	619	570	670	36.2	2.95	[22]
17	dye	polymer	PMMA	film	412	300	600	513	450	650	28.0	2.95	[23]
18	QD	polymer	EVA	film	300	300	550	515	450	550	35.9	3.08	[19]

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19	QD	polymer	OSTE	bulk	480	400	660	622	550	700	15.7	3.20	[24]
20	dye	polymer	PMMA	bulk	590	350	650	604	500	750	64.0	3.30	[25]
21	CD	polymer	PDMS	film	300	200	675	518	425	675	44.0	3.71	[21]
22	CD	polymer	siloxane	film	358	225	425	441	400	600	94.0	3.90	[26]
23	dye	polymer	PMMA	bulk	370	300	500	509	480	750	68.0	4.40	[27]
24	Ln	glass	glass	bulk	583	400	950	1060	900	1100	32.0	4.50	[28]
25	CD	polymer	PVP	film	490	300	520	510	450	650	80.5	4.80	[29]
26	dye			film	575	300	610	620	570	720	3.0	4.80	[30]
27	CD	polymer	PDMS	film	280	200	650	448	425	650	56.0	5.10	[21]
28	CD	polymer	p(MMA-NIPAm)	bulk	340	300	500	520	400	700	6.0	5.84	[31]
29	QD	polymer	lauryl methacrylate and ethylene glycol dimethacrylate	bulk	620	560	775	885	690	1240	45.0	6.10	[32]
30	dye	polymer	PMMA	bulk	520	400	550	580	500	700	99.0	7.50	[33]
31	dye	polymer	polycarbonate	bulk	576	300	610	651	575	800	96.0	8.10	[34]
32	dye	polymer	polycarbonate	bulk	576	300	610	658	575	800	96.0	8.10	[34]
33	dye	polymer	PS	bulk	576	300	610	646	575	800	96.0	8.40	[34]
34	dye	polymer	PMMA	film	533	400	625	656	550	850	77.0	8.40	[35]
35	CD	polymer	PMMA	bulk	470	300	550	535	450	650	9.6	9.30	[36]
36	dye	polymer	PMMA	bulk	380	300	500	509	480	700		10.40	[37]
37	dye	hybrid	d-U(600)	bulk	570	300	600	650	350	750	80.0	14.50	[38]
38	dye	polymer	PDMS	bulk	446	250	500	553	500	800	89.5	31.30	[39]
4	dye	polymer	PDMS	bulk	575	300	610	620	570	720	100.0	33.00	[11]
39	dye	polymer	(simulated n=1.5)	bulk	575	400	610	620	570	700	96.0	37.70	[40]
40	QD	polymer	PLMA-co-EGDA	bulk	450	400	600	580	550	650	86.0	48.00	[41]
41	dye	hybrid	PDMS	bulk	521	450	550	539	500	700	93.0	54.00	[42]
42	Ln	polymer	PMMA	bulk	300	300	350	613	570	710	86.0	60.00	[43]

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43	dye	polymer	PMMA	bulk	575	300	600	600	550	750	100.0	60.00	[43]
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SI.3. Dataset structure

After pre-processing, the dataset is structured with 19 input features and 1 output feature. These features are derived from the 11 characterization parameters obtained from the repository dataset. The first 19 attributes are used as input in the input layer of the model, while the last attribute represents the experimental values for the optical power efficiency and serves as a benchmark to compare against the model's output. After pre-processing, the dataset is organized with 19 features. These features are derived from the 11 classifiers obtained from the repository dataset. Among these, the first four are categorical classifiers, which have been transformed into 12 features using one-hot encoding. The remaining seven classifiers are numerical. The initial 19 attributes are utilized as input in the input layer of the model. The final attribute represents the experimental values for the optical power efficiency and serves as a reference point for comparing against the model's output.

Input features																			Output feature
Materials												Optical							
Optical centre type			Material host type			Material host chemical composition			Processing method			A_p	A_{min}	A_{max}	E_p	E_{min}	E_{max}	η_{yield}	η_{opt}
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20

Figure S1. Scheme of the dataset structure, highlighting the connections between the features employed as inputs in the ANN and the classifiers within the dataset.

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